03-621 Week 11

Advanced Quantative Genetics

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Finding the Genes

What is the actual genetic architecture of complex traits?

- **Ronald Fisher's Infinitesimal Model (1918)**
- Large number of genes, with common alleles each contributing a small additive effect.

Does this model hold up in the face of modern GWAS evidence, especially for common threshold disease traits? How important are rare alleles with large effect? How important are non-additive effects?

Attempts to Account for a Greater Proportion of Heritability

We want to account for:

- Type II error
- Infinitesimal hypothesis

Methods:

- The basic concept is to combine the effects of **all** of the SNPs in the study (not just significant SNPs). Is any h^2 really missing?
- Several software packages available. (e.g., GCTA: as random effects by a mixed linear model)

The SNP genotypes are used to calculate genetic similarities among unrelated individuals and this is regressed on phenotypic similarities. Strongly criticized as severely overestimating or underestimateing h^2 due to multiple flaws and sensitivities to data imperfections (e.g., population structure due to admixture)

False Positives due to Population Admixture

When two populations with different trait prevalence intermix, failure to take mixed genetic background into account can lead to false positives in GWAS.

- The reason: any allele(s) that distinguish the two populations will appear to be associated with the trait

From Mapping to Identifying Genes

General strategies

- Follow-up mapping and association studies at higher resolution
- Extend analysis to rare variants in delimited regions
- Establish causation

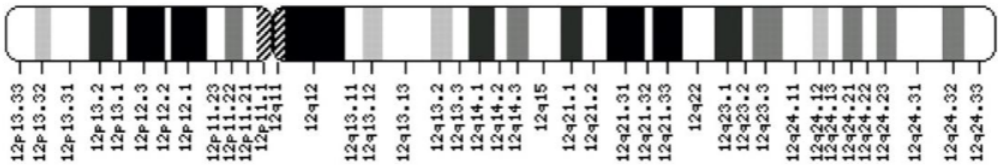
Some special approaches: Sometimes useful for monogenic traits and some complex traits

- Exome sequencing
- Cytological Mapping using chromosome abnormalities

Route to Pinpointing the Responsible Gene and Mutations

1. Extend Association Studies to tracing inheritance in families
   - Known common and rare alleles from databases
   - New rare alleles identified by sequencing the candidate genes in patients and controls
2. Predict effect of mutation on gene function
   - From first principles (null alleles, frameshifts, missense mutations)
   - From knowledge or modeling of structure and functional domains
3. Verify effect on gene function
   - In vitro assays of purified wild type and mutant proteins
   - In vivo assays in cell lines or model organisms
4. Verify effect on phenotype
   - Gene knockout or replacement in cell lines or model organisms

Cytological Mapping Using Chromosomal Abnormalities

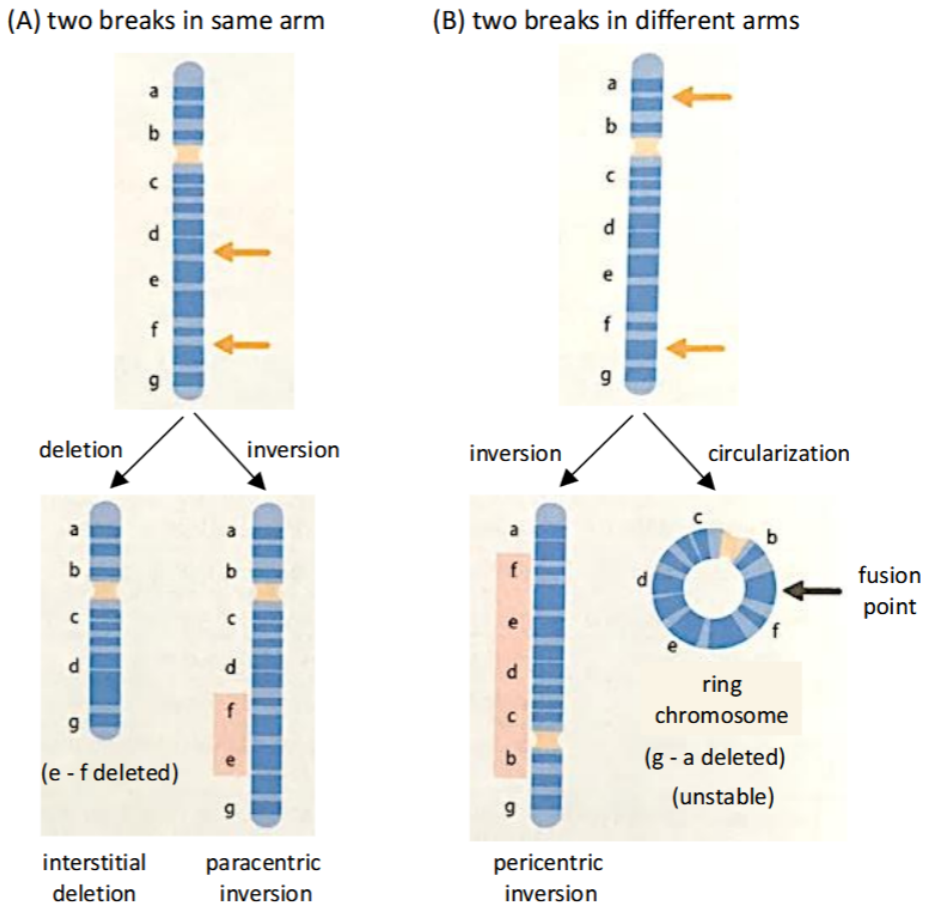


The banding patterns of chromosomes allow us to identify the location of breakpoints in chromosome abnormalities that may **inactive** or **inappropriately activate** a gene. Useful for specific cases of:

- Rare disease traits with insufficient number of informative pedigrees
- Highly deleterious dominant traits
  - Not transmitted
  - All cases are new sporadic mutations

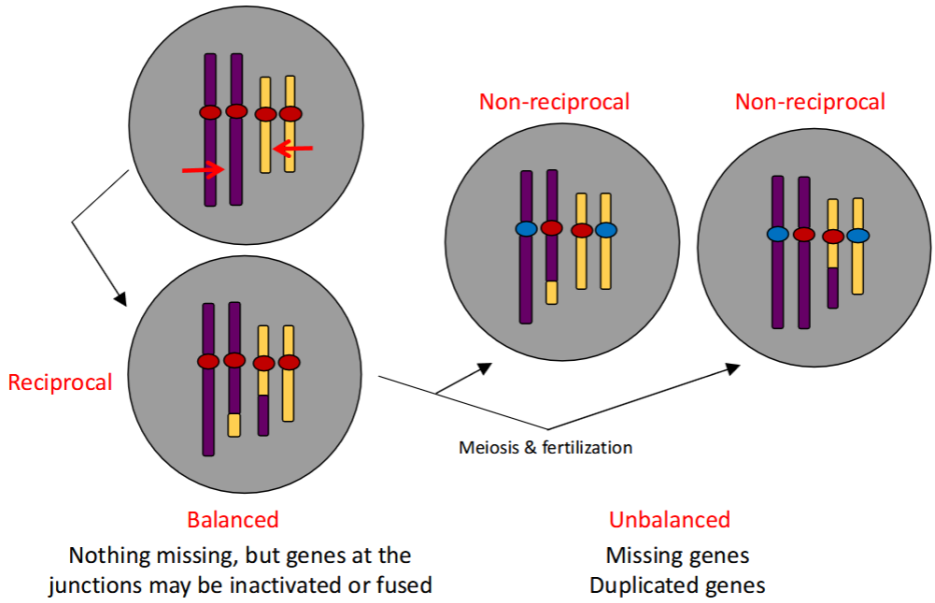
Abnormalities 1: Inversions and Deletions

Genes at the junctions may be **inactivated** or **fused**. Fusing is a common mechnanism of activation of cancer-promoting genes.



Abnormalities 2: Translocations

Result from breaks on different chromosomes.



Examples of Copy Number Polymorphisms (CNPs) Associated with Complex Diseases

CNP Allele	Disease
Deletion upstream of *IRGM* (innate immune response)	Crohn's disease
Low-copy of repeat in Lipoprotein A gene	Coronary artery disease
Extra copies of the β-defensin multigene family (antimicrobial peptides)	Psoriasis
Deletion of 1 copy of the complement *C4* gene	Systemic lupus erythematosus
Reduced copies of *FCGR3A* (Fc portion of IgG)	

Examples of Loci that Exhibit Copy Number Variation in Autism Spectrum Disorder

Locus Size	Genes	Location	Pathogenic Allele	Frequency in ASD (%)
0.7 Mb	30 genes	16p11.2	deletions and duplications	0.8
$\sim$ 1 Mb	*PTCHD1AS* *PTCHD1*	Xp22.1	deletions affecting *PTCHD1AS* (upstream antisense ncRNA)	0.5
Variable	*NRXN1*	2p16.3	mostly deletions	0.4
1.4 Mb	22 genes	7q11.2	duplication	0.2
2.5 Mb	56 genes	22q11.2	deletions and duplications	0.2
1.5 Mb	14 genes	1q21.1	duplications	0.2
Variable	*shank2*	11q13.3	deletions	0.1

Note that the copy number variations each account for only a **small** percentage of ASD cases

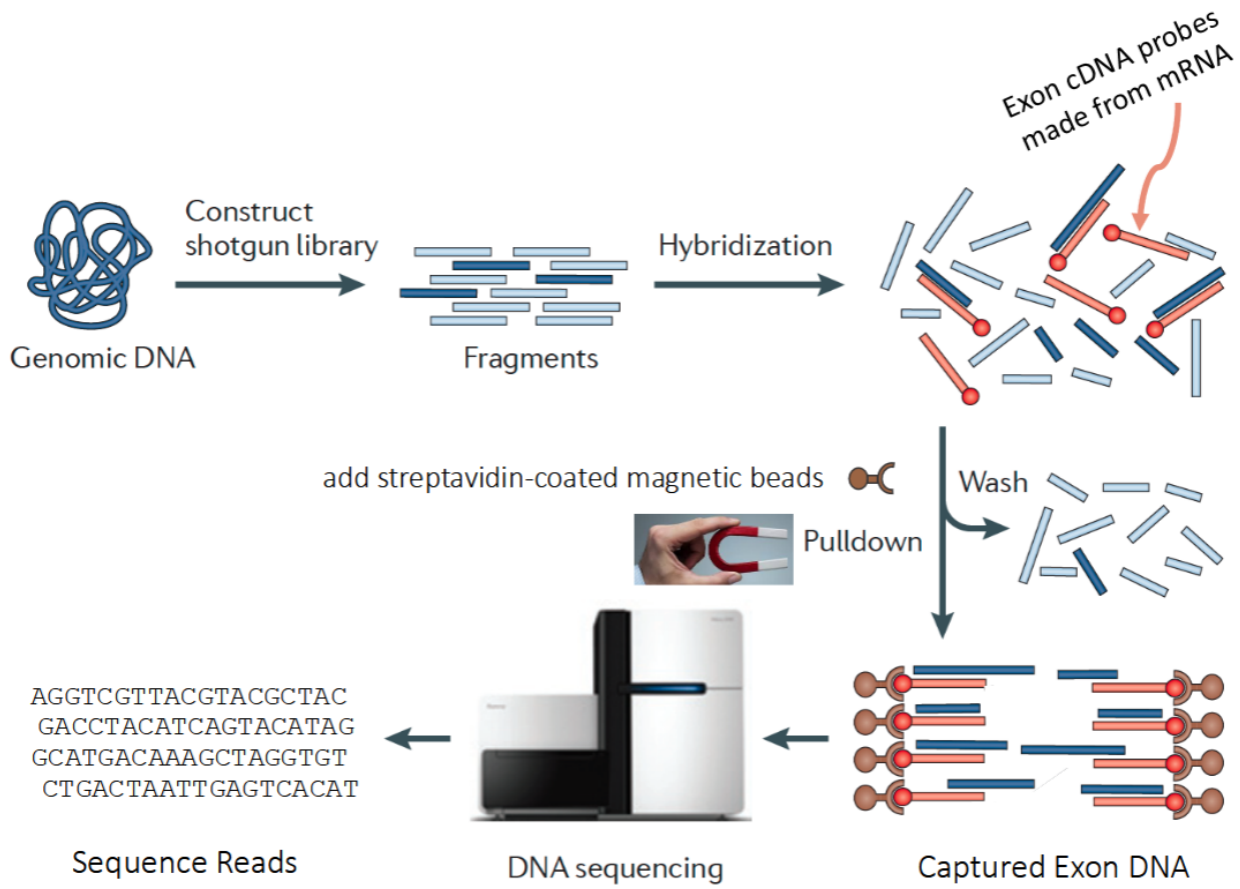
Exome Sequencing

Also useful for single-gene disorders that are:

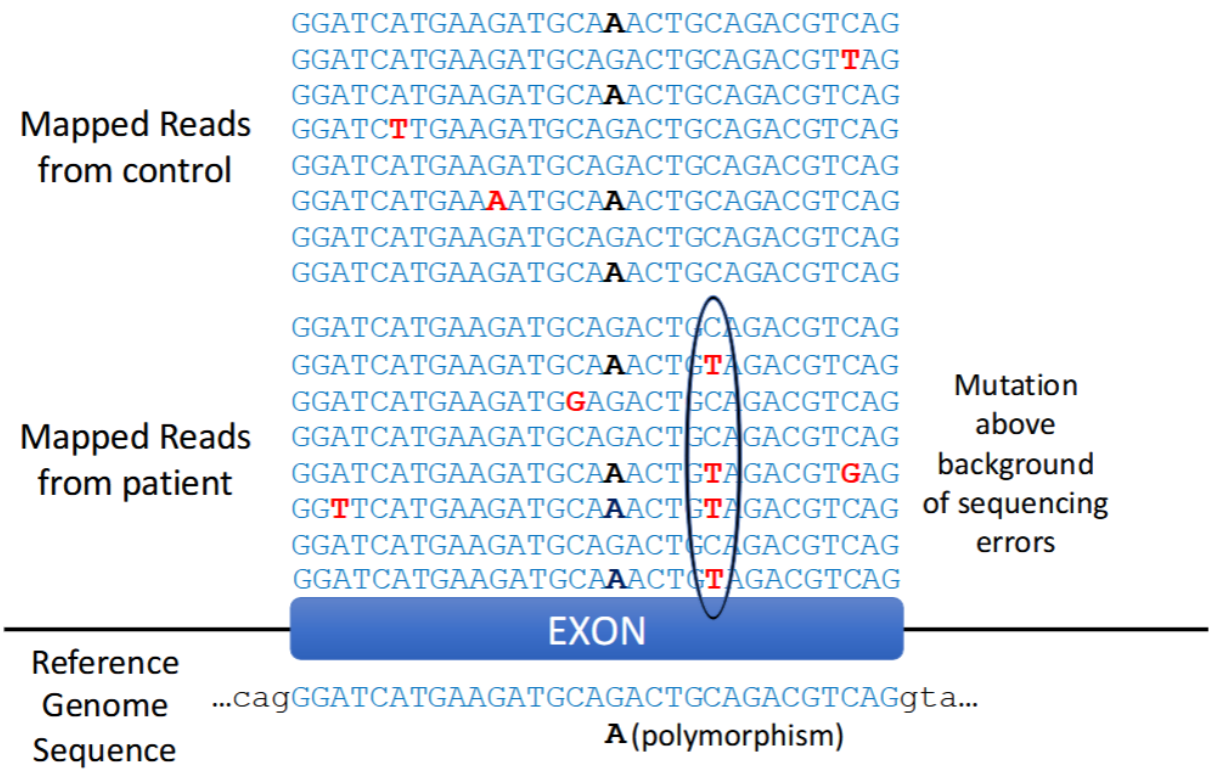
- Very rare disease traits with an insufficient number of informative pedigrees for mapping
- Highly deleterious dominant traits
  - not transmitted
  - therefore, all cases are *new* sporadic mutations
  - therefore, cannot be mapped

Exome sequencing method

Targets sequencing to exons.



Exon-seq reads are aligned to the genome and compared to identify mutations (e.g., rare dominant)



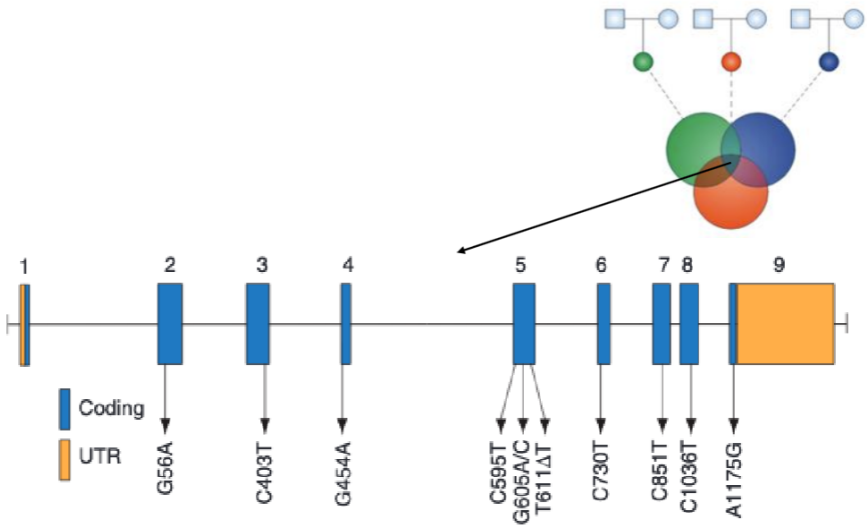
Examples of Successful Exome Sequencing

Very rare recessive or severe dominant monogenic diseases

Disease	Inheritance	Type of Disorder	Exomes Sequenced From
Miller Syndrome	AR	congenital disorders of development	unrelated sporadic cases
Kabuki Syndrome	AD		
Schinzel-Giedion Syndrome	AD		
Osteogenesis imperfecta Type VI	AR	connective tissue disorder	affected siblings born to consanguineous parents
Spastic paraplegia 30	AD	early-onset neuromuscular disorder	

Exome Sequencing of Siblings to Find the Genetic Basis of the Miller Syndrome

1. Find variants in patients
2. Filter for homozygous alleles (trait is recessive)
3. Remove common variants found in many people.
4. Compare to mutations in other unrelated affected children



11 different LOF mutations in the *DHODH* gene found among 6 families with high rate of Miller syntdrome

- *DHODH*: Biosynthesis of pyrimidine nucleotides: C, T, U for DNA, RNA

Other Uses of Exome-Seq

Exome-seq is also useful for phenotypes with **extremely heterogeneous genetics**. e.g., Autosomal Recessive Intellectual Disabilities (ARID)

- Note the involvement in shared functional networks:
  - additional proof of relevance
  - clues to identify additional relevant genes
  - clues to underlying molecular pathology

Limitations of Exome Sequencing

- Relevant mutations are **not necessarily** in Exons!
- Relevant mutations are **not necessarily** in protein-coding genes at all!
- Generally, not useful for complex traits